NEXUSGEAR

PRESENTATION 02

Specification & Prototype

E-Commerce Project

A curated home for high-performance gaming peripherals and ergonomic desk setups.

Popa-Ion Andrei

Frontend Lead & Customer Experience

UI/UX • client flows • AJAX • responsive design

Șerban Alexandru-Cristian-George

Backend Lead & System Integrations

Layered .NET 10 • 3NF data • Stripe • tests

ACADEMIC YEAR 2026–2027 • P2 / WEEK 7

THE PROTOTYPE

From product discovery to a tested specification.



KEYBOARDS

8 PRODUCTS



GAMING MICE

8 PRODUCTS



DESKMATS

8 PRODUCTS



ERGONOMIC GAMING CHAIRS

6 PRODUCTS

30 deterministic products • 4 categories
53 passing xUnit tests • 4 bonus features implemented

Context, store niche & progress since P1

A specialist buying journey for enthusiast peripherals and ergonomic desk setups.

THE NICHE

One focused store.
Four connected categories.

TARGET CUSTOMERS

Gamers	Precision & performance
Programmers	Comfort for long sessions
Remote workers	Healthier desk setups

CATALOG • DETERMINISTIC DBSEEDER

30 products / 4 categories
8 keyboards • 8 mice • 8 desk mats • 6 chairs

P1 → P2 | WHAT CHANGED

01 — EXECUTABLE .NET 10 MONOLITH

NexusGear + NexusGear.Tests • Repository Wrapper • Unit of Work • Service Layer

02 — EXPANDED 3NF DATA MODEL

11 relational tables • 14 foreign keys • M:N HAS_ADDRESS

03 — AHEAD-OF-SCHEDULE PROTOTYPE

A1, A2, A3, A4, A11, A12 plus B1, B2, B3, B7

53 / 53 TESTS PASS — Automated xUnit suite • single dotnet test command

Frozen specification — mandatory requirements A1–A6

P2 status is distinguished from the requirements scheduled for P3.

A1 PRODUCT CATALOG

IMPLEMENTED

Paginated listing (6 or 12/page; always ≤24), category views by unique Slug, product details with live stock/specs and deterministic SQLite seed data.

A2 BASIC SEARCH

IMPLEMENTED

Case-insensitive title + keyword search via EF Core LINQ; empty-state view includes one-click filter reset.

A3 SHOPPING CART

IMPLEMENTED

CartService uses HTTP Session for guests and IN_CART for users; merges on login; server computes subtotals and enforces stock limits.

A4 USER ACCOUNTS

IMPLEMENTED

Cookie authentication with BCrypt.Net-Next; customer profile supports multiple saved addresses through M:N HAS_ADDRESS.

A5 CHECKOUT FLOW

SPECIFIED • P3 / WEEK 8

Four steps: address, Standard Courier $15 / Express Next-Day $35, discount, review and NXG-2026-XXXX confirmation.

A6 SANDBOX PAYMENT

SPECIFIED • P3 / WEEK 9

Order starts Unpaid; only a cryptographically verified checkout.session.completed webhook at /api/webhook/stripe changes it to Paid.

Frozen specification — mandatory requirements A7–A12

Six further acceptance areas; implemented items are shown separately from P3 scope.

A7 STOCK MANAGEMENT

SPECIFIED • P3 / WEEK 9

Atomic SQLite transaction on verified payment; EF Core RowVersion concurrency prevents negative stock; cancellation restores inventory.

A8 ORDER HISTORY

SPECIFIED • P3 / WEEK 10

Authenticated OrdersController dashboard with chronology, status, shipping details and historical CONTAINS UnitPrice snapshots.

A9 ADMINISTRATION PANEL

SPECIFIED • P3 / WEEK 10

Custom AdminController, Admin role, product/category CRUD, local image upload and controlled order transitions.

A10 BASIC SECURITY

IMPLEMENTED & ENFORCED

ModelState validation, antiforgery on state-changing POSTs, parameterized EF queries and server-side Admin role guards (403).

A11 RESPONSIVE DESIGN

IMPLEMENTED

Mobile-first Bootstrap 5.3; customer-facing pages remain usable without horizontal overflow at 360 px.

A12 AUTOMATED TESTS

IMPLEMENTED • 53 TESTS

NexusGear.Tests runs on isolated In-Memory SQLite; catalog, cart, auth, wishlist, reviews, preview/error coverage.

Frozen specification — bonus features B1–B7

Seven optional features extend the frozen specification alongside mandatory requirements A1–A12.

B1 — FILTERED SEARCH — IMPLEMENTED

Category + price + stock filters, price/date sorting; state preserved in URL.

B2 — RECOMMENDATIONS — IMPLEMENTED

Frequently Bought Together from CONTAINS co-occurrence; Similar Products by category and price tier.

B3 — REVIEWS & RATINGS — IMPLEMENTED

Verified-buyer gate; average rating on cards; Admin IsHidden moderation.

B4 — SALES DASHBOARD — PLANNED P3

Date range, Chart.js daily revenue, Top 5 products and AOV.

B5 — DISCOUNT CODES — SEEDED / P3

NEXUS10 and SETUP50; expiry, max uses and server-side validation.

B6 — TRANSACTIONAL EMAIL — PLANNED P3

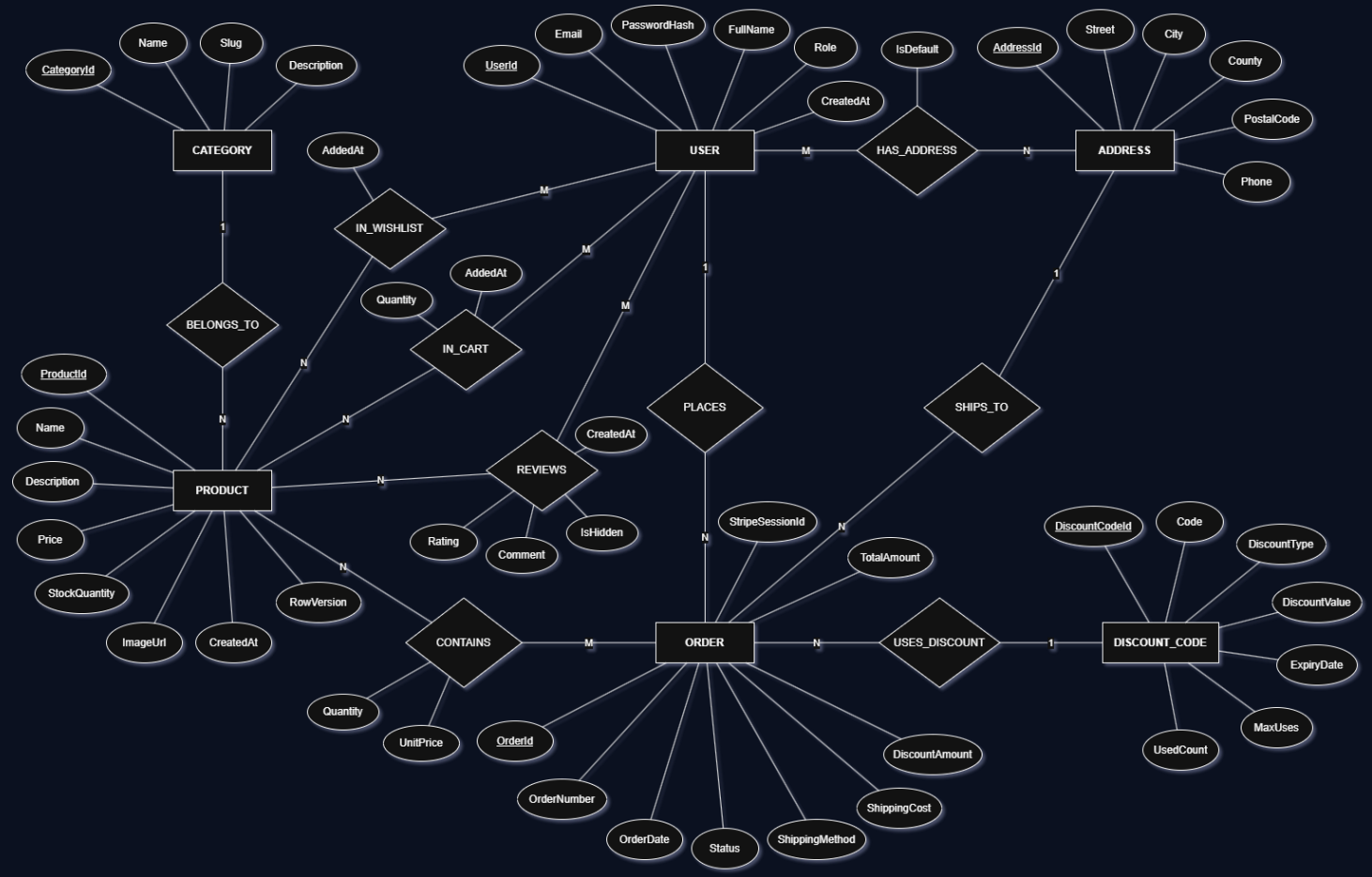
Mailtrap SMTP HTML receipt after verified Stripe webhook.

B7 — WISHLIST — IMPLEMENTED

IN_WISHLIST persistence, navbar count and Move to Cart.

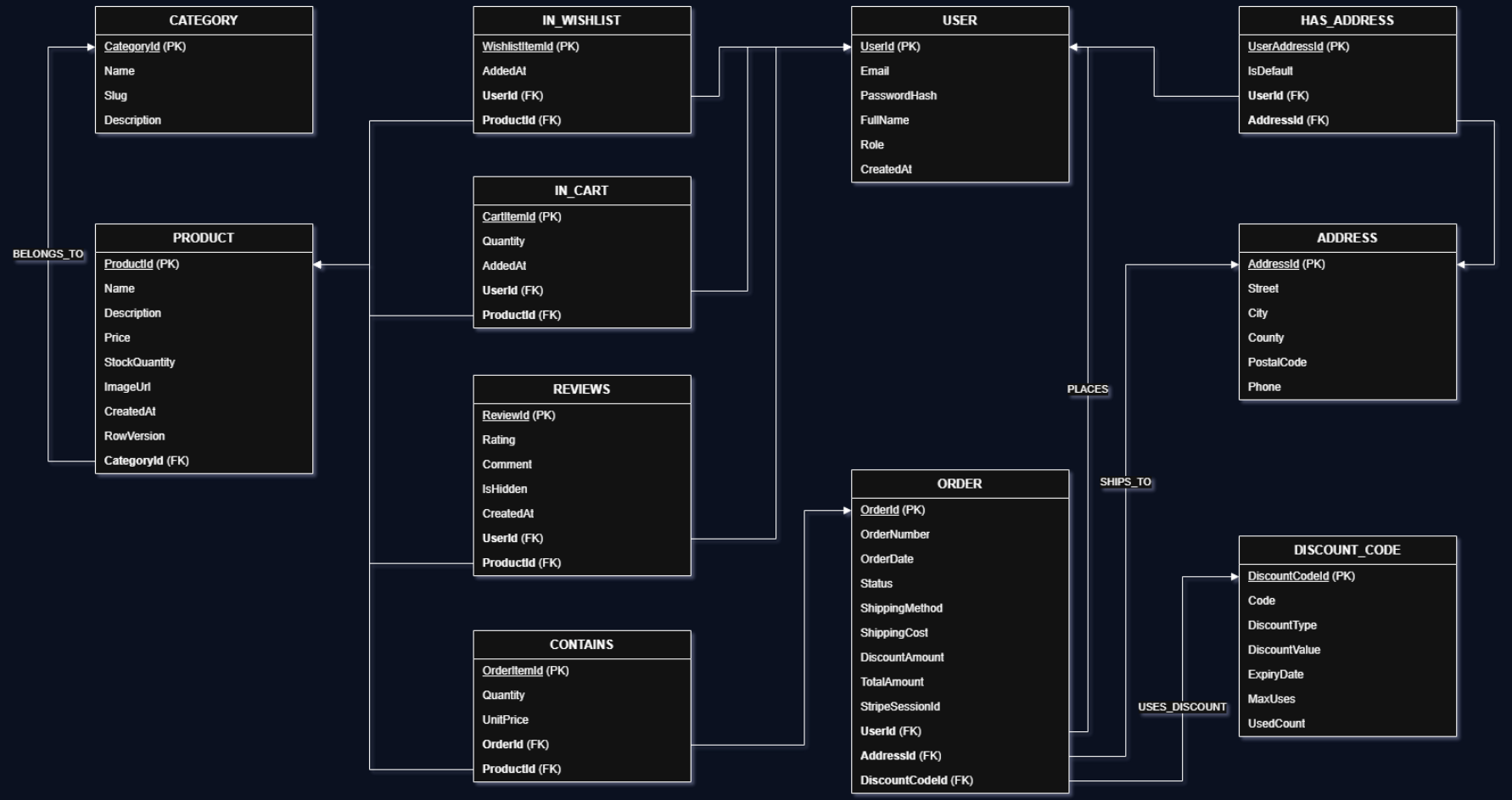
Conceptual data model — Chen ERD

Six core entities • four 1:N relationships • five associative M:N relationships.



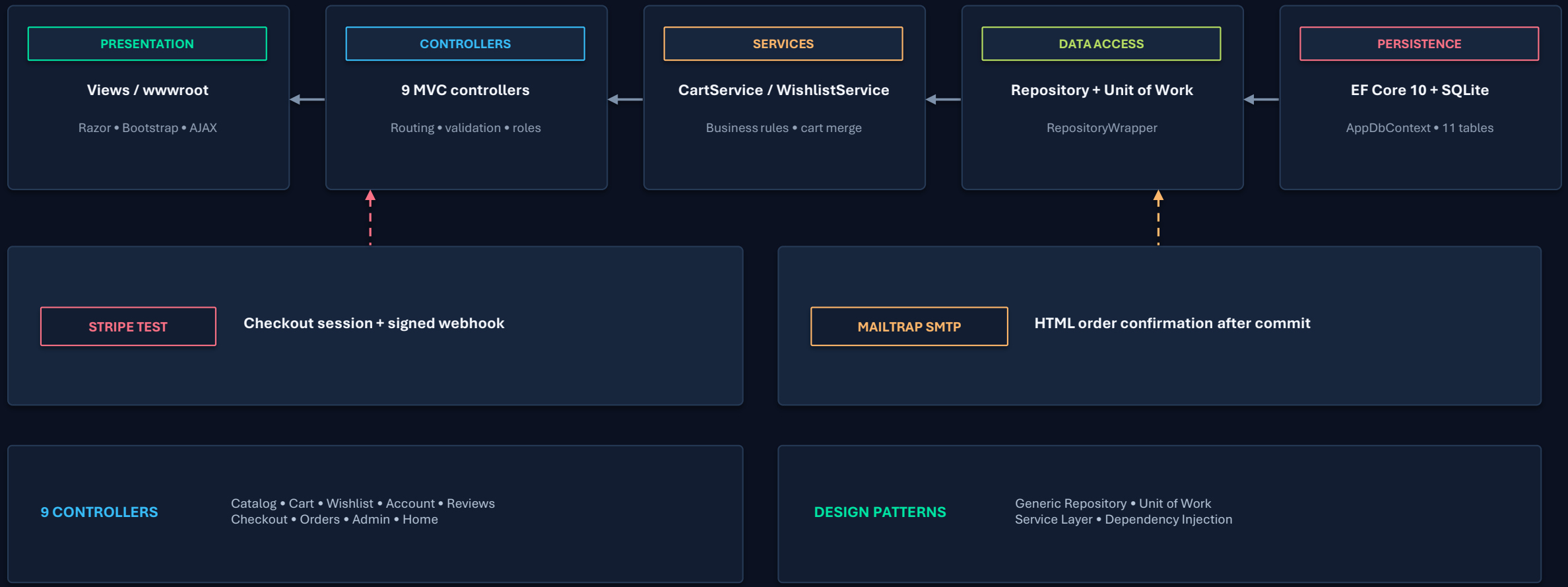
Physical relational model — 11 tables in 3NF

Junction tables keep the same names as the five M:N diamonds in the ERD.



High-level system architecture & design patterns

Layered monolith • dependency injection • Repository • Unit of Work • Service Layer.



API endpoints & access control

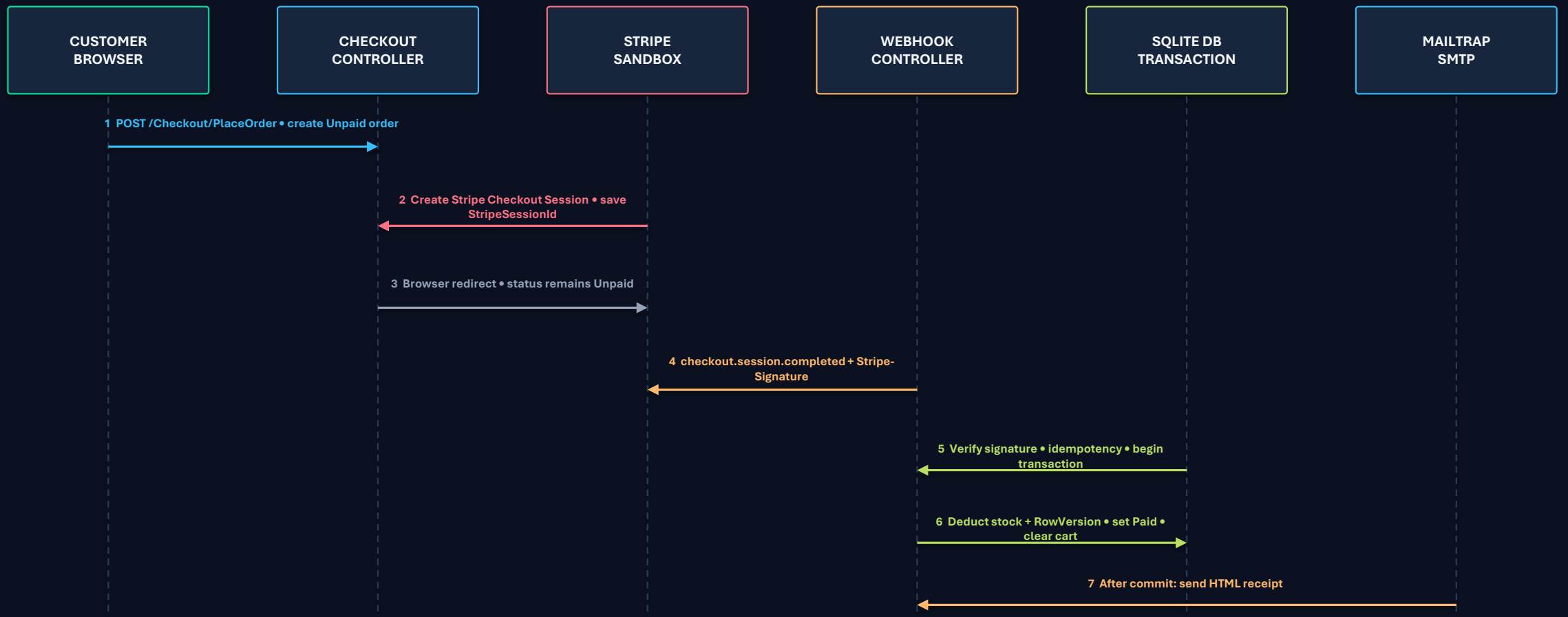
Complete route catalogue • access is enforced server-side, not only by the UI.

METHOD	ROUTE	ACCESS / PURPOSE
GET	/	Public Home / storefront
GET	/Catalog	Public Catalog + URL filters • A1/A2/B1
GET	/Catalog/Details/{id}	Public Details + B2/B3
GET	/Cart	Public / Customer Session or IN_CART • A3
POST	/Cart/Add	Public / Customer Stock validation; AJAX/POST • A3
POST	/Cart/UpdateQuantity	Public / Customer Quantity cap / remove • A3
POST	/Cart/Remove	Public / Customer Recalculate totals • A3
GET / POST	/Wishlist / Toggle	Customer Wishlist via AJAX • B7
POST	/Wishlist/MoveToCart	Customer Move saved item to cart • B7
GET / POST	/Account/Register / Login	Public BCrypt + cookie auth • A4

METHOD	ROUTE	ACCESS / PURPOSE
POST	/Account/Logout	Customer / Admin Cookie logout + CSRF • A4/A10
GET	/Account/Profile	Customer HAS_ADDRESS • A4
POST	/Account/AddAddress	Customer Create/link ADDRESS • A4
POST	/Account/SetDefaultAddress	Customer Set IsDefault • A4
POST	/Account/DeleteAddress	Customer Unlink HAS_ADDRESS • A4
POST	/Reviews/Create	Verified buyer Paid/Delivered check • B3
POST	/Reviews/ToggleVisibility	Admin Review moderation • A9/B3
GET / POST	/Checkout / PlaceOrder	Customer 4-step flow + discounts • A5/B5
POST	/api/webhook/stripe	Stripe signature Verify + stock + Paid • A6/A7/B6
GET / POST	/Orders / Details/{id}	Customer Own order history • A8
GET / POST	/Admin / Products/* / Orders/*	Admin role CRUD, image upload, dashboard • A9/A10/B4/B5

Critical flow — payment confirmation & stock concurrency

The browser redirect never marks an order Paid; only a verified server-to-server webhook does.



UI wireframes — catalog & product detail

Conceptual wireframes based on the specified live prototype screens.

nexusgear.ro / Catalog

NEXUSGEAR Search gear... Categories Wishlist Cart

FILTERS B1

Search title

Category

Min / Max price

In stock only

Sort by price/date

APPLY

MECHANICAL KEYBOARDS • 30 PRODUCTS



NexusForge K-75

$129

ADD TO CART



NexusTitan K-75

$149

ADD TO CART



NexusCore TKL

$99

ADD TO CART

AJAX Add to Cart • stock badges • rating summary • URL pagination

nexusgear.ro / Catalog / Details / 1

PRODUCT DETAIL



NexusForge K-75

★★★★★ 4.8 rating

$129.00

IN STOCK

ADD TO CART

TECH SPECS & DESCRIPTION

Frequently Bought Together • B2

Verified Customer Reviews • B3

Admin moderation: IsHidden toggle

UI wireframes — cart & checkout wizard

Cart interactions are specified as AJAX or standard POST; checkout is a P3 build milestone.

nexusgear.ro / Cart

YOUR CART • FREE SHIPPING PROGRESS

Spend $292 more to unlock free shipping



NexusForge K-75

$129

− 1 +



NexusPulse M-40

$79

− 1 +

ORDER SUMMARY

Subtotal	$208
Shipping	$15
Discount	-$10
Total	$213

CHECKOUT

CartService • server totals • stock-limit enforcement • guest-to-user merge

nexusgear.ro / Checkout

CHECKOUT 1 ADDRESS → 2 DELIVERY → 3 DISCOUNT → 4 REVIEW

STEP 1 • DELIVERY ADDRESS

Default address • IsDefault = true

STEP 2 • SHIPPING METHOD

Standard
$15.00

Express
$35.00

STEP 3 • PROMO CODE

NEXUS10 / SETUP50

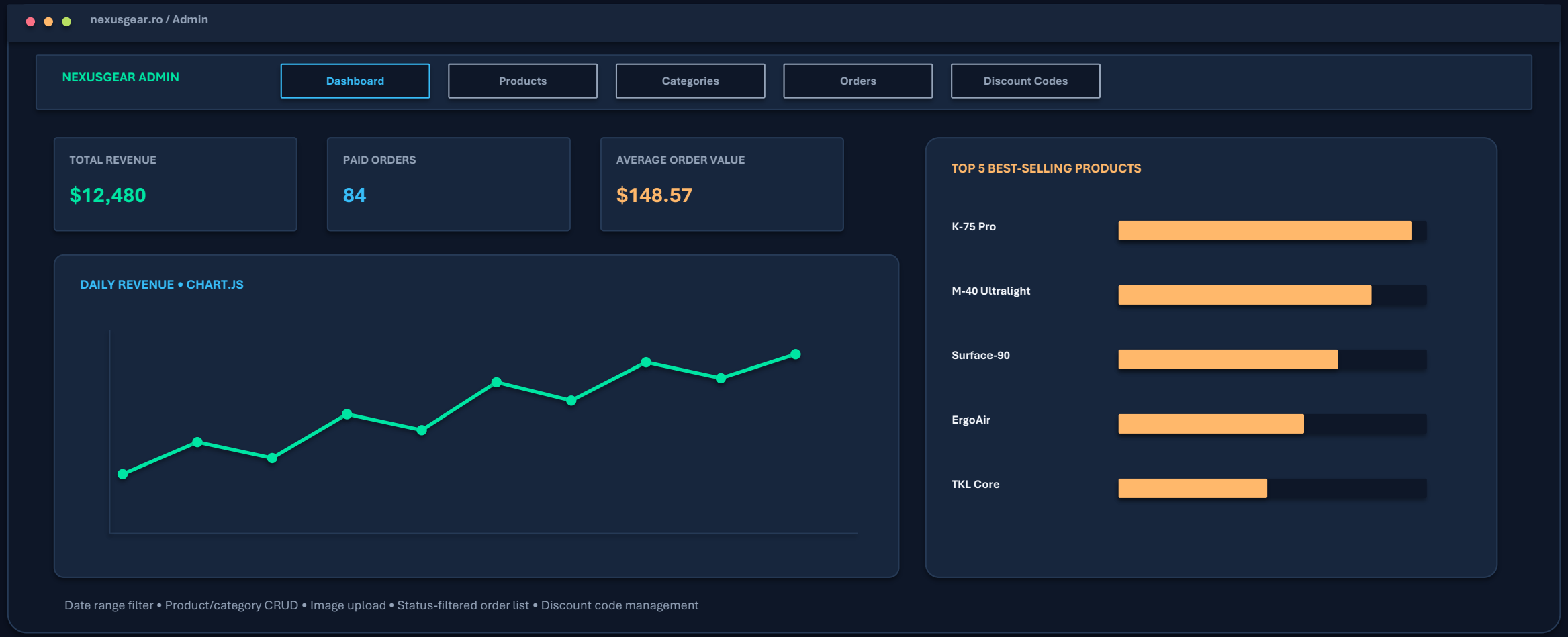
ORDER REVIEW

Subtotal
Shipping
Discount
Final total

PAY WITH STRIPE

UI wireframe — admin panel & sales dashboard

Custom back-office • catalog management • order fulfillment • analytics and discounts.



Functional prototype demo & 53 passing xUnit tests

P2 prototype starts locally, seeds deterministic data and demonstrates the implemented customer journey.

LIVE DEMO FLOW

01	START	dotnet run --project NexusGear
02	BROWSE	Catalog • categories • filters • recommendations
03	CART	Guest add-to-cart • server totals • stock cap
04	SIGN IN	customer@nexusgear.ro • merge session cart
05	PROFILE	Saved address book • wishlist • verified review

Demo accounts are seeded for local testing only.

NEXUSGEAR.TESTS • 53 / 53 PASSED

CatalogTests.cs	14	Pagination • search • B1 • B2
CartTests.cs	13	Totals • stock • guest merge
AccountAndReviewTests.cs	12	BCrypt • addresses • B3 moderation
WishlistTests.cs	8	IN_WISHLIST • Move to Cart
PreviewAndErrorTests.cs	6	Admin role • preview routes • errors

`dotnet test` Isolated In-Memory SQLite • 5 suites

Local seeded customer: customer@nexusgear.ro
Local seeded admin: admin@nexusgear.ro

Specification freeze & Stage 3 roadmap

P2 freezes the approved scope; implementation and live validation continue in Weeks 8–12.

P2 READINESS

6 / 12 mandatory requirements complete

A1 A2 A3 A4 A11 A12	Implemented and tested at P2
B1 B2 B3 B7	Four bonus features implemented
B4 B5 B6	Planned / seeded for P3

RoadmapPreview routes keep Stage 3 entry points reachable without HTTP 404 errors.

STAGE 3 • WEEKS 8–12

W8	A5 • Checkout wizard + B5 discount application	Assigned: Popa-Ion Andrei
W9	A6 • Stripe webhook + A7 inventory concurrency + B6 Mailtrap	Assigned: Șerban Alexandru-Cristian-George
W10	A9 • Admin CRUD + B4 dashboard + A8 order history	Assigned: Both team members
W11–12	A12 webhook/concurrency tests • A10/A11 audit • backup demo ≤5 min • P3 tag	Assigned: Both team members

LIVE DEMO

NEXUSGEAR • P2 / SPECIFICATION FROZEN